

Chapter 13

Lecture 13: Properties and Criteria of compactness.

Definition 13.0.1. A collection $\mathcal{A}$ of subsets of a space X is called "a cover" or "a covering" if the union of the elements of $\mathcal{A}$ is X ; it is called an open covering if all these sets are open.

Definition 13.0.2. X is called a compact space if every open covering $\mathcal{A}$ contains a finite subcollection which also covers X .

Remark Being compact depends on the topology: For example, consider X with the antidiscrete topology, i.e., $\mathcal{T} = \{\emptyset, X\}$. Then X is compact.

Example 13.0.1. $\mathbb{R}$ with the standard topology is not compact. Consider the open covering $\mathcal{A} = \{(-n, n) | n \in \mathbb{N}\}$.

Example 13.0.2. $(0, 1]$ with the standard topology is not compact. Consider the open covering $\mathcal{A} = \{(1/n, 1] | n \in \mathbb{N}\}$.

Example 13.0.3. • $\{0\} \cup \{\frac{1}{n} | n \in \mathbb{Z}^+\} \subseteq \mathbb{R}$ is compact because $\{0\}$ is the limit point and any open covering must contain an open set containing 0, which covers all but finite points.

• $\{\frac{1}{n} | n \in \mathbb{Z}^+\}$ is not compact. Consider the open covering $\mathcal{A} = \{(1/n - \epsilon, 1/n + \epsilon) | n \in \mathbb{N}\}$. (Similarly, for infinite X with discrete topology, X is not compact.)

Definition 13.0.3. $Y \subseteq X$ is compact if every open covering of Y by sets open in X contains a finite subcollection covering Y .

Lemma 13.0.1. $Y \subseteq X$ is compact if and only if every open covering of Y by sets open in Y contains a finite subcollection covering Y .

Proof

Trivial.

Theorem 13.0.1. Every closed subspace of a compact space is compact.

Proof Let X be a compact space and let Y be a closed subspace of X . Let $\mathcal{A}$ be an open covering of Y by sets open in X . Since Y is closed, $X \setminus Y$ is open in X . Thus, $\mathcal{A} \cup \{X \setminus Y\}$ is an open covering of X . By the compactness of X , there exists a finite subcollection $\mathcal{A}' \subseteq \mathcal{A}$ such that $\mathcal{A}' \cup \{X \setminus Y\}$ covers X . Therefore, $\mathcal{A}'$ covers Y . Hence, Y is compact.

Theorem 13.0.2. Every compact subspace of a Hausdorff space is closed.

Proof

It suffices to show that for any $x \in X \setminus Y$, there exists an open set U containing x such that $U \cap Y = \emptyset$. For each $y \in Y$, since X is Hausdorff, there exist disjoint open sets U_y and V_y such that $x \in U_y$ and $y \in V_y$. Then $\{V_y | y \in Y\}$ is an open covering of Y . By the compactness of Y , there exists a finite subcollection $\{V_{y_1}, V_{y_2}, \dots, V_{y_n}\}$ that covers Y . Let

$$U = \bigcap_{i=1}^n U_{y_i} \tag{13.1}$$

Then U is an open set containing x . Moreover

$$U \cap Y = \left(\bigcap_{i=1}^n U_{y_i} \right) \cap Y = \emptyset \quad (13.2)$$

Thus, $U \cap Y = \emptyset$. Hence, Y is closed. ■

Lemma 13.0.2. *Y is a compact subspace of Hausdorff space X . And $x \in X \setminus Y$. Then there exist open sets U and V such that $x \in U$, $Y \subseteq V$ and $U \cap V = \emptyset$.*

Proof Proved inside the proof of the theorem.

Theorem 13.0.3. *The image of a compact set under a continuous map is compact.*

Proof Let $f : X \rightarrow Y$ be a continuous map. Consider any open covering $\mathcal{A}$ of $f(X)$ by sets open in Y . Then $\{f^{-1}(U) | U \in \mathcal{A}\}$ is an open covering of X by sets open in X . By the compactness of X , there exists a finite subcollection $\{f^{-1}(U_1), f^{-1}(U_2), \dots, f^{-1}(U_n)\}$ that covers X . Thus, $\{U_1, U_2, \dots, U_n\}$ is a finite subcollection of $\mathcal{A}$ that covers $f(X)$. Hence, $f(X)$ is compact. ■

Theorem 13.0.4. *Let $f : X \rightarrow Y$ be a bijective continuous map. If X is compact and Y is Hausdorff, then f is a homeomorphism.*

Proof

We just combine the above theorems.

It suffices to show that images of closed sets under f are closed. Consider any closed set D in X . By the theorem, D is compact. Thus, by the previous theorem, $f(D)$ is compact. Since Y is Hausdorff, by another theorem, $f(D)$ is closed. Hence, f is a homeomorphism. ■

Lemma 13.0.3 (tube lemma). *Let $x_0 \in X$, suppose $\{x_0\} \times Y$ is covered by open sets W_i in $X \times Y$. Then one can choose a finite subcovering $W_1, W_2, \dots, W_n$ of $\{x_0\} \times Y$ and find an open neighbor U of x_0 such that $U \times Y \subseteq \bigcup_{i=1}^n W_i$.*

Proof One may assume that $W_i = U_i \times V_i$ where U_i is open in X and V_i is open in Y . Then $\{V_i\}$ is an open covering of Y . By the compactness of Y , there exists a finite subcollection $\{V_1, V_2, \dots, V_n\}$ that covers Y .

To be done.

Theorem 13.0.5. *The product of finitely many compact spaces is compact.*

Proof It is enough to prove for two sets. Let X and Y be compact spaces.

We can cover $X \times Y$ by finitely many tubes by the tube lemma. And each tube can be covered by finitely many open sets. Thus $X \times Y$ is compact.

To be done.

Definition 13.0.4. *A collection $\mathcal{C}$ of sets has the finite intersection property if for any finite subcollection $\{C_1, C_2, \dots, C_n\} \subseteq \mathcal{C}$, we have*

$$C_1 \cap C_2 \cap \dots \cap C_n \neq \emptyset \quad (13.3)$$

Theorem 13.0.6. *X is a topological space. Then X is compact if and only if for every collection $\mathcal{C}$ of closed sets in X with the finite intersection property, we have*

$$\bigcap_{C \in \mathcal{C}} C \neq \emptyset \quad (13.4)$$

Proof

Let $\mathcal{A}$ be a collection of open sets in X . Then $\mathcal{C} = \{X \setminus A | A \in \mathcal{A}\}$ is a collection of closed sets in X .

Then $\mathcal{A}$ is an open covering of X if and only if $\bigcap_{C \in \mathcal{C}} C = \emptyset$.

Then a finite subcollection of $\mathcal{A}$ covers X if and only if the corresponding finite subcollection of $\mathcal{C}$ has empty intersection.

Now the theorem follows directly. ■